

Chapter 5: Orthogonal Linear Operator

Saturday, 23 March 2024

* Defn
s.1 A real $n \times n$ matrix A is called orthogonal if $A^T A = I$

* Defn
s.2 Orthogonal group $O_n = \{A_{n \times n} \text{ s.t. } A^T A = I\}$

* Defn
s.3 Special Orthogonal group $SO_n = \{A_{n \times n} \text{ s.t. } A^T A = I \text{ and } \det(A) = 1\}$

* Defn
s.4 The dot product of $u, v \in \mathbb{R}^n$ is defined as
$$(u, v) = x_1 y_1 + x_2 y_2 + x_3 y_3 + \dots + x_n y_n$$
 where
 $x = \{x_1, x_2, \dots, x_n\}^T, y = \{y_1, y_2, \dots, y_n\}^T$ are coordinate vectors of u, v w.r.t the standard basis
[Symmetric bilinear operator 9.2]

* Defn
s.5 The length of vector $u \in \mathbb{R}^n$ is defined as
$$\|u\|^2 = (u, u)$$

* Defn
s.6 $u, v \in \mathbb{R}^n$ are called orthogonal if $(u, v) = 0$

* Defn
s.7 A basis $\mathcal{B} = (v_1, \dots, v_n)$ of $\mathbb{R}^n$ is called orthonormal if it consists of orthogonal unit vectors i.e. if $v_i, v_j \in \mathcal{B}$ then
$$(v_i, v_j) = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{if } i \neq j \end{cases}$$

* Defn
s.8 A linear operator T on $\mathbb{R}^n$ is called orthogonal if it preserves the dot product between all the vectors, i.e.
$$(T(u), T(v)) = (u, v) \quad \forall u, v \in \mathbb{R}^n$$

* Lemma
s.9 If a linear operator on $\mathbb{R}^n$ preserves the length of vectors, then it is an orthogonal linear operator.

Chapter 5: SO3 & Rotation SO2 and SO3

Saturday, 23 March 2024

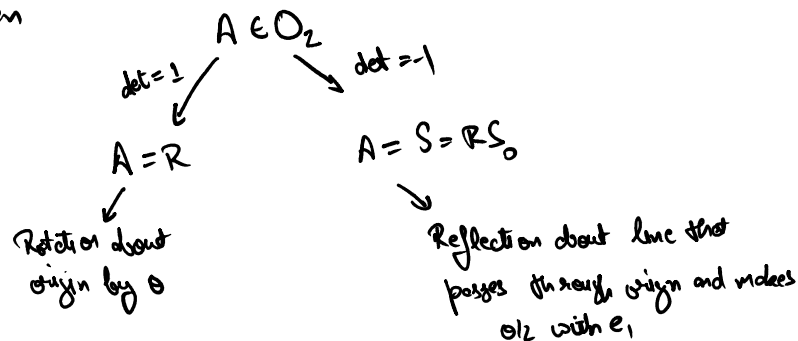
* Lemma S.10 The transformation matrix of an orthogonal linear operator $\mathbb{R}^2$ w.r.t the standard basis is orthogonal

* Defn: S.11 A linear operator T on $\mathbb{R}^2$ is called a reflection if it has orthogonal eigen vector v_1 and v_2 with eigen values 1 and -1

* Defn: S.12 $R = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$, $S_0 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, $S = RS_0 = \begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{bmatrix}$

* Lemma S.13 If $x=R$ $y=S_0$ then $S=xy = y(x^{-1})$ $\swarrow$ Rotation by $-\theta$

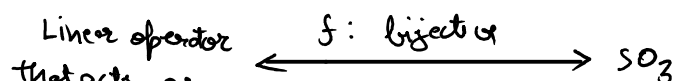
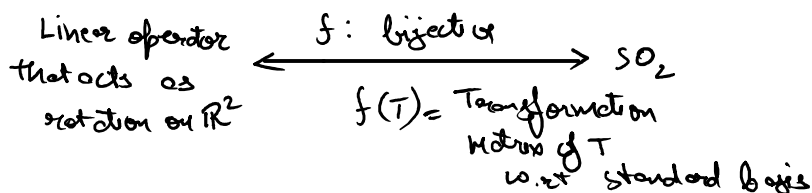
* Lemma S.14 Let A be the transformation matrix of an orthogonal linear operator T w.r.t. standard basis then



* Defn S.15 An orthogonal linear operator on $\mathbb{R}^3$ is called a rotation if
 ① It fixes a unit vector u , referred to as the pole
 ② It rotates the 2 dimensional subspace orthogonal to u

* Lemma S.16 If $A \in SO_3$ Then 1 is a diagonal value of A .

* Lemma S.17



Linear operator
 that acts as
 rotation on $\mathbb{R}^3$

f : bijective

SO_3

$f(T) =$ Transformation
 matrix of T
 w.r.t standard basis

Chapter 5: Cayley-Hamilton Theorem

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* Lemma 5.17 Let $M \in SO_3$ with the spin (u, α) . Let $B \in SO_3$ and $u' = Bu$ then $M' = BMB^T$ represents rotation with spin (u', α)

the coefficient of the leading term is 1

* Lemma 5.18 Let $p_k(t)$ be a sequence of monic polynomial of $\deg \leq n$ and $p(t)$ be a monic polynomial of $\deg n$. Let $\alpha_{k,1}, \alpha_{k,2}, \dots, \alpha_{k,n}$ and $\alpha_1, \alpha_2, \dots, \alpha_n$ be the roots of these polynomials

a) If $\alpha_{k,i} \rightarrow \alpha_i$ then $p_k \rightarrow p$

b) If $p_k \rightarrow p$ then $\alpha_{k,i}$ can be numbered s.t $\alpha_{k,i} \rightarrow \alpha_i$ for $i=1, \dots, n$

* Lemma 5.19 If $A_{n \times n}$ is a complex matrix then $\exists$ a sequence of matrices A_k s.t

① $A_k \rightarrow A$

② A_k has distinct roots

Proof Hint: Use 1.14, 4.24

* Cayley Hamilton Theorem: 5.20 If $p(t) = t^n + c_{n-1}t^{n-1} + c_{n-2}t^{n-2} + \dots + c_0$ is the characteristic polynomial of a $A_{n \times n}$ complex matrix, then

$$p(A) = A^n + c_{n-1}A^{n-1} + \dots + c_1A + c_0I = O_{n \times n}$$

* Defn 5.21 Vector valued function

$$x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix}$$

$$x'(t) = \begin{bmatrix} x_1'(t) \\ x_2'(t) \\ \vdots \\ x_n'(t) \end{bmatrix}$$

Matrix value function

$$A(t) = \begin{bmatrix} a_{11}(t) & \dots & a_{1n}(t) \\ a_{21}(t) & a_{22}(t) & \dots \\ \vdots & \ddots & \vdots \\ a_{n1}(t) & \dots & a_{nn}(t) \end{bmatrix}$$

$$A'(t) = \begin{bmatrix} a'_{11}(t) & \dots & a'_{1n}(t) \\ \vdots & \ddots & \vdots \\ a'_{n1}(t) & \dots & a'_{nn}(t) \end{bmatrix}$$

* Defn 5.22 If $A_1, A_2, \dots, A_k$ are differential matrix valued function then

$$\frac{d}{dt}(A_1 A_2 \dots A_k) = \sum_{i=1}^k A_1 A_2 \dots A_{i-1} \frac{d(A_i)}{dt} A_{i+1} \dots A_k$$

Chapter 5: General Solution

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* Defn
5.23 Exponential of a complex number $a = x + si$ is defined by
$$e^a = e^{x+si} = e^x (\cos(s) + i \sin(s))$$

* Defn
5.24 Dependent variable and its derivatives appear only to power 1 and not multiplied together

(The highest derivative is first order)

Linear, First order, constant coefficient differential equation
$$\frac{dx}{dt} + P(x)x = Q(x) \quad \text{--- } \underline{\underline{5.25}}$$

If $Q(x) = 0$ then the equation is called Homogeneous linear, first order constant coefficient differential equation

$$\frac{dx}{dt} + P(x)x = 0 \quad \text{--- } \underline{\underline{5.26}}$$

- * Every single solution of 5.25 is called a particular solution
- * Solutions to 5.26 are called homogeneous solutions

* Lemma
5.27 For 5.25 General solution = Homogeneous solution + Particular solution

Proof :- Let $x_1(t), x_2(t)$ solve 5.1 then

$$\frac{d}{dt}(x_2(t) - x_1(t)) + P(x)(x_2(t) - x_1(t)) = Q(x) - Q(x) = 0$$

$$x_2(t) - x_1(t) = x_h(t)$$

homogeneous solution

$$x_2(t) = x_h(t) + x_1(t)$$

Chapter 5: Matrix Exponential

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* Lemma 5.28 Let $A_{n \times n}$ be a matrix s.t. $\Lambda = P^{-1}AP$ is diagonal with entries $\lambda_1, \lambda_2, \dots, \lambda_n$ for some invertible matrix P . Then the general solution to $\frac{dx}{dt} = Ax$ is $x = P\tilde{x}$ where $\tilde{x} = (c_1 e^{\lambda_1 t}, c_2 e^{\lambda_2 t}, \dots, c_n e^{\lambda_n t})^T$ solves $\frac{d\tilde{x}}{dt} = \Lambda \tilde{x}$

* Lemma 5.29 Let $x(t)$ be a complex valued solution of $\frac{dx}{dt} = \underbrace{A}_{n \times n \text{ real matrix}} x$ then real and complex part solve the same equation.

* Defn 5.30 If A is a $n \times n$ matrix then
$$e^A = I + \frac{A}{1!} + \frac{A^2}{2!} + \frac{A^3}{3!} + \dots \quad \text{5.3}$$

* Theorem: 5.31 (a) The series 5.2 converge absolutely and uniformly on a bounded set of complex matrix

(b) e^{tA} is differentiable function of t and $\frac{d}{dt} e^{tA} = A e^{tA}$

(c) If $AB = BA$ then $e^{A+B} = e^A e^B$

* Lemma 5.32 e^A where $A_{n \times n}$ complex matrix, is invertible

* Theorem 5.23 The columns of matrix e^{tA} form basis for the space of solutions of $\frac{dx}{dt} = Ax$
Hint $\frac{d}{dt} (e^{-tA} x(t)) = 0 \quad \therefore x(t) = e^{tA} c$

Chapter 6: Isometry Group M_n

* Defn 6.1 Let P, Q are points in $\mathbb{R}^n$ and u, v are vector pointing to them respectively.
Distance b/w P and $Q = |\overrightarrow{PQ}| = |v - u|$ defined in 5.5

* Defn 6.2 Isometry: $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$
(distance preserving m.p.s) s.t if $u, v \in \mathbb{R}^n$, $|f(u) - f(v)| = |u - v|$

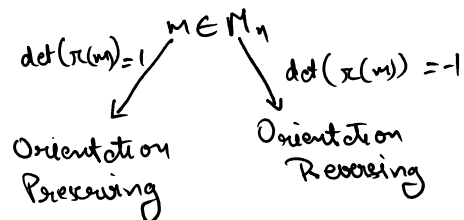
* Lemma 6.3 An isometry that fixes the origin $\iff$ Orthogonal linear operator

* Lemma 6.4 All isometries of $\mathbb{R}^n$ together form a group M_n

* Lemma 6.5 If $m \in M_n$, then $\exists$ a unique translation isometry t_a and an orthogonal isometry φ s.t $m = t_a \varphi$

* Lemma 6.6 Let t_a be the translation isometry and φ be the orthogonal linear operator then $\varphi t_a = t_{\varphi(a)} \varphi$

* Lemma 6.7 Group Homomorphism $M_n \xrightarrow{\pi} O_n \xrightarrow{\det} \pm 1$



$$\pi(t_a \varphi) \rightarrow \varphi$$

* Lemma 6.8 In $\mathbb{R}^n$, let C and C' be two Coordinate systems associated with an isometry $n = t_a \varphi$
 $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be an isometry with formula m, m' w.r.t coordinates C and C' then $m' = n^{-1} m n$

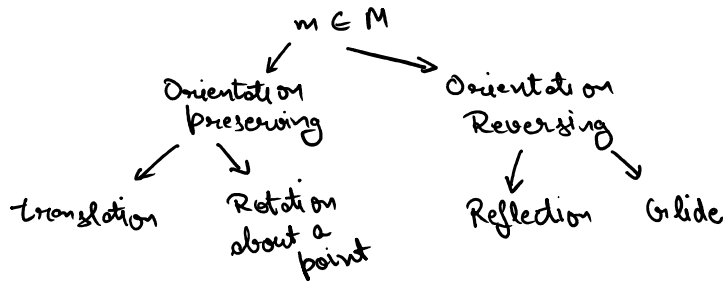
* Defn 6.9 M : Isometry group of $\mathbb{R}^n$
 $\hookrightarrow$ is generated by $\underbrace{t_a}_{\text{translation}}$, $\underbrace{\rho_\theta}_{\text{rotation about origin by } \theta}$, $\underbrace{\pi}_{\text{reflection about } e_1 \text{ axis}}$

Chapter 6: Finite Subgroups of O_2

Saturday, 23 March 2024

* Lemma

6.10



* Defn The subgroup Γ of $(\mathbb{R}, +)$ is called discrete if

6.11

$$\exists \varepsilon > 0 \text{ s.t. } \forall a \in \Gamma \quad \text{abs}(a) > \varepsilon$$

* Lemma

6.12

If $\Gamma \subset \mathbb{R}^+$ is discrete then $\Gamma = a\mathbb{Z}$ where $a \in \mathbb{R}$

* Defn $G \subset M$ is called discrete if $\exists \varepsilon > 0$ s.t if $\begin{cases} ta \in G, a \neq 0 \text{ then } |a| > \varepsilon \\ t_0 g_0 \in G, g_0 \neq 0 \text{ then } |g_0| > \varepsilon \end{cases}$

6.13

Hint: No need to add a condition specific to $m \in G$ that is a glide or reflection, as it is a translation

* Lemma

6.14

$$G \subset O_2 \text{ has finite order} \iff G \subset O_2 \text{ is discrete}$$

C_n Cyclic group generated by g_0 where $\theta = \frac{2\pi}{n}$, $n \in \mathbb{Z}$
 D_n Dihedral group generated by α and β_0 where $\theta = \frac{2\pi}{n}$, $n \in \mathbb{Z}$

Fixed Point

* Theorem: If $G \subset M$ has finite order then G fixes a point and if coordinates are selected appropriately G becomes C_n or D_n

6.15

* Defn

6.16

The translation group L of $G \subset M$ is

$$L = \{ v \in \mathbb{R}^2 \mid t_v \in G \}$$

* Lemma

6.17

Let L be a discrete subgroup of the group $(\mathbb{R}^2, +)$ then

a) A bounded region contains only finitely many points

* Lemma

6.17

Let L be a discrete subgroup of the group $(\mathbb{K}, +)$ then

- a) A bounded region contains only finitely many points
- b) If L is non trivial then it contains a non zero vector of minimal length.

Chapter 6: Crystallographic Restriction

Saturday, 23 March 2024

* Lemma 6.18 Let $B = (u, v)$ be the basis of $(\mathbb{R}^2, +)$. Let L be the lattice of integer combinations of B . Then every $v \in (\mathbb{R}^2, +)$ can be uniquely written as $v = x + v_0$ where $x \in L$ and v_0 in $\pi'(B)$.
 $\pi'(B)$: 11 gram with vertices $(0, u, v, u+v)$ and edges $(u, u+v), (v, u+v)$ missing

* Lemma 6.19 If $L \subset (\mathbb{R}^2, +)$ is discrete

- $L = \{0\}$
- $L = a\mathbb{Z}$ where $a \in \mathbb{R}^2$
- $L = a\mathbb{Z} + b\mathbb{Z}$ where $a, b \in \mathbb{R}^2$

* Defn 6.20 The point group $\bar{G}$ of GCM is
 $\bar{G} = \{g \in O_2 \text{ st } \exists t_a g \in G\}$

* Theorem 6.21 Let GCM is discrete, $a \in L$, $\bar{g} \in \bar{G}$ and $a' = \bar{g}(a) \Rightarrow a' \in L$.
 In other words, for a discrete group, its point group maps its translation group to itself

* Lemma 6.22 Crystallographic restrictions:- Let L be a discrete subgroup of $(\mathbb{R}^2, +)$, and let $H \subset O_2$ be a subgroup of the symmetries of L . If L is non trivial, then

- * every rotation in H has the order 1, 2, 3, 4 or 6
- * H is one of the group C_n or D_n with $n = 1, 2, 3, 4$ or 6

* Caution: The point group $\bar{G}$ and the translation group L of the discrete subgroup GCM doesn't completely identify the group

* Lemma 6.23 The crystallographic restriction limits the types of discrete GCM to 17.

* Lemma The constraint of crystallographic restriction can be extended to 3D to conclude there

6.24 * Lemma The concept of crystallographic restriction can be expanded to 3D to conclude there are 230 crystallographic groups

Chapter 6: Group Action

Saturday, 23 March 2024

* Defn 6.25 Group Operation: Let G be group and S be a set, then the group operation of G on S is map $f: G \times S \rightarrow S$

st ① $e.s = s \quad \forall s \in S$, e identity elements of G

② $g.(g'.s) = (gg').s \quad \forall g, g' \in G, \forall s \in S$

* Defn 6.26 Orbit O_s of $s \in S$ is defined as

$$O_s = \{s' \in S \text{ st } g.s = s' \text{ where } g \in G\}$$

* Defn 6.27 Stabilizer G_s of $s \in S$ is defined as

$$G_s = \{g \in G \text{ st } g.s = s\}$$

* Defn 6.28 If S consists of only one orbit the group operation is called transitive

* Lemma 6.29 Let $s \in S$, G operates on S and $H \subset G$ is the stabilizer of s then

① If $a, b \in G$ st $a.s = b.s$ then $a^{-1}b \in H$, i.e. $a = bH$

② If $s' = a.s$ then the stabilizer of s' is $H' = aHa^{-1}$

* Lemma 6.30 Let $H \subset G$, then the operation of G on the set of left cosets G/H is transitive $G/[H]$ is H . Also if $[u] \in G/H$ then $G[u] = gHg^{-1}$ where $g \in G$

* Theorem 6.31 Let S be a set on which G operates, and let $s \in S$ and O_s and H are its orbit and stabilizer. Then there exist a bijective map

$$\varepsilon: G/H \rightarrow O_s$$

$$\varepsilon([aH]) = a.s$$

* Lemma 6.32 Counting Formula: $|G| = |H| |G/H|$
 $= |H| |O_s|$

* Lemma 6.33 The relation of belonging to an orbit is an equivalence relation $\Rightarrow$ the orbits partition the set S .

* Lemma
6.33

The relation of belonging to an orbit is an equivalence relation $\Rightarrow$ the orbits partition the set S .

$$|S| = |O_1| + |O_2| + \dots + |O_k|$$

Chapter 6: Permutation representation

Saturday, 23 March 2024

* Lemma 6.34 A permutation representation of a group G that operates on a set S is the group homomorphism $\varphi: G \rightarrow \text{Perm}(S)$

* Defn 6.35 $\text{Perm}(S)$ = Group of permutations of set S
 $\text{Perm}(S) \neq$ Permutation representation

* Lemma 6.36 $\left[\begin{array}{c} \text{Operations of } G \\ \text{on set } S \end{array} \right] \xrightleftharpoons[\text{correspondence}]{\text{bijective}} \left[\varphi: G \rightarrow \text{Perm}(S) \right]$

* Defn 6.37 A group operation is called faithful if the only element $g \in G$ st $g \cdot s = s \quad \forall s \in S$ is identity, i.e. the kernel of the permutation representation is $\{e\}$.

* Theorem 6.38 Let G be a finite subgroup of SO_3 . Let P represent the set of poles of elements of G . Then G operates on P as a group action

* Lemma 6.39 A finite subgroup of SO_3 is one of the following

- ① C_k : Cyclic group
- ② D_k : Dihedral group
- ③ T : Tetrahedral group of 12 rotational symmetries of tetrahedron
- ④ O : Octahedral group of 24 rotational symmetries of a cube or a octahedron
- ⑤ I : Icosahedral group of 60 rotational symmetries of dodecahedron or icosahedron

Hint: Use $G \curvearrowright P \xrightarrow{\text{group action}} \overline{P}$ set of poles

and count spins to get

$$\sum \left(1 - \frac{1}{n_p}\right) = 2 - \frac{2}{N}$$

→ Lemma 6.40 There exists only five regular polyhedra

Chapter 7: p Group

Saturday, 23 March 2024

Defn 7.1 Conjugation: the operation of G on itself is defined by
 $(g, x) \mapsto g x g^{-1}$

Defn 7.2 Center of a group G is
 $Z = \{ x \in G \text{ s.t. } x g = g x \ \forall g \in G \}$

Defn 7.3 Centralizer of an element x of a group G
 $Z(x) = \{ g \in G \text{ s.t. } g x = x g \}$

Lemma 7.4 $Z(x)$ contains $\begin{array}{l} \rightarrow x \\ \rightarrow Z \\ \rightarrow \text{Maybe some other elements} \end{array}$

Lemma 7.5 $|G| = |Z(x)| \cdot |C(x)|$
Conjugacy class of x

Lemma 7.6 $|G| = |G_1| + |G_2| + \dots + |G_k|$

Lemma 7.7 If $x \in Z$ then its conjugacy class contains only x itself.

Defn 7.8 p Group: A group is called p group if its order is positive power of a prime number

Lemma 7.9 The center of a p group is non trivial

Lemma 7.10 Let G , a p group, acts on a finite set S of order n not divisible by p . Then S contains at least one element which is fixed by all elements of G

*Lemma 7.11 Every group of order p^2 is abelian

* Lemma ^{7.11} Every group of order p^2 is abelian

* Lemma 7.12 A group of order p^2 is either cyclic or isomorphic to the product of two cyclic groups of order p

Chapter 7: Conjugation in S_n

Saturday, 23 March 2024

* Defn 7.13 A group G is called Simple if it has no proper normal subgroups.

* Lemma 7.14 If a normal subgroup N contains an element x , then N also contains the conjugacy class of x .

* Lemma 7.15 $p, p' \in S_n$ are conjugates if and only if their cycle decompositions have the same order.

* Lemma 7.16 For $n \geq 3$, A_n is generated by 3-cycles.

* Lemma 7.17 For $n \geq 5$, the 3-cycles form a single conjugacy class in A_n .

* Theorem 7.18 For $n \geq 5$, $A_n \leq S_n$ is a simple group.

* Defn 7.19 Let $H \leq G$. Consider conjugation.
Orbit of $H =$ conjugate subgroups of form gHg^{-1} where $g \in G$.
Stabilizer of $H = N(H) = \{g \in G \text{ s.t. } gHg^{-1} = H\}$

* Lemma 7.20 $\begin{cases} H \triangleleft N(H) \\ H \triangleleft G \text{ if } N(H) = G \end{cases}$

* Lemma 7.21 If $|S| = n = p^c m$, where m is not divisible by p . Then
Let $U = \{ \text{subset of } S \text{ of order } p^c \}$

$|U| = N = \binom{n}{p^c}$ is not divisible by p

* Theorem 7.22 If G is a group, U is its subset and H is stabilizer of U .
Then $|H|$ divides $|G|$ and also $|H|$ divides $|U|$.

Hint: $H \times U \rightarrow U$

$$|U| = |O_{u_1}| + |O_{u_2}| + \dots + \overbrace{|O_{u_k}|}^{\text{Right cosets } [\because \text{same size}]}$$

$0 > |H|$

Chapter 7: Sylow's p subgroup

Saturday, 23 March 2024

* Defn 1.23 If $|G| = p^c m$ where $c \geq 1$ and p doesn't divide m . A Sylow p subgroup is the subgroup of G of order p^c

* First Theorem 1.24 If $|G| = p^c m$, where $c \geq 1$ and p doesn't divide m . Then there exists Sylow's p subgroups of G . Hint $G \times S \rightarrow \sum$ of G of order p^c

* Second Theorem 1.25 If G contains an p subgroup K . Then $K \subset gHg^{-1}$, where H is an Sylow's p subgroup
 All the Sylow's p subgroups are conjugates of each other

Hint:- left multiplication $G \times G/H \xrightarrow{\text{transitive}} G/H$ n : Sylow p group

$c = [H] \in G/H$ with stabilizer n

$K \times G/H \rightarrow G/H$ K : p group

$$K \subset Hc' = gHg^{-1}$$

$|G/H|$ not divisible by $|K|$

$\therefore \exists c' \in G/H$ stabilized by K

Also since $G \times G/H \rightarrow G/H$ is transitive $c' = gC$ and $Hc' = gHg^{-1}$

* Third Theorem 1.26 If s is the number of Sylow's p subgroups. Then

① s divides m

② $s = kp + 1$ where $k \in \mathbb{Z}$

Hint $G \times S \xrightarrow[\text{Transitive}]{\text{conjugation}} S$

s : Set of Sylow p subgroup

$[H] \in S$ Sylow p subgroup

$$|G| = |N(H)| |O_H|$$

$$= |N(H)| |S|$$

$\therefore s$ divides m

$$H \subset N(H)$$

$$\therefore p^c \leq |N(H)| \leq p^c m$$

conjugation $H \times S \rightarrow S$

$\therefore S$ divides n

conjugation
 $H \times S \rightarrow S$

show only $(1) \in S$ has orbit of order 1

Chapter 8: Hermitian Form

Saturday, 23 March 2024

* Defn
8.1 Symmetric Matrix
 $A^T = A$

Orthogonal Matrix
 $A^T A = I$

* Defn
8.2 Symmetric Bilinear Form
 $\langle, \rangle : V \times V \rightarrow \mathbb{R}$ s.t

- ① Linear in first variable
- ② " " second "
- ③ $\langle v, w \rangle = \langle w, v \rangle \quad \forall v, w \in V$

* Defn
8.3

Matrix of form: A where
 $a_{ij} = \langle v_i, v_j \rangle$
 $[B] = (v_1, v_2, \dots, v_n)$
basis

* Lemma
8.4 Matrix of the form is
Symmetric

* Lemma
8.5 $\langle v, w \rangle = x^T A y$

where x and y are
coordinate vectors of v, w w.r.t $[B]$

* Lemma
8.6

If $[B]$ and $[B']$ are two basis of V s.t $[B'] = [B]P$ and
 A, A' are matrices of $\langle, \rangle$ w.r.t to $[B]$ and $[B']$ then

$$A' = P^T A P$$

Normal Matrix
 $A^* A = A A^*$

Hermitian Matrix
 $A^* = A$
conjugate transpose

Unitary Matrix
 $P^* P = I$

Hermitian form

$$\langle, \rangle : V \times V \rightarrow \mathbb{C}$$

- ① Conjugate linear in 1st variable
- ② Linear in 2nd variable
- ③ $\langle v, w \rangle = \overline{\langle w, v \rangle} \quad \forall v, w \in V$

Matrix of the form is
Hermitian

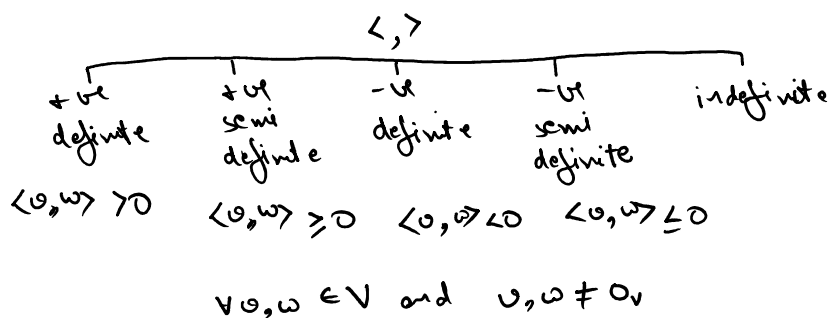
$$\langle v, w \rangle = x^* A y$$

$$A' = Q^* A P$$

Chapter 8: Orthogonality

Saturday, 23 March 2024

* Lemma
8.1



* Defn
8.8 $v \perp w$ if $\langle v, w \rangle = 0$

* Defn
8.9 Orthogonal space $W^\perp$ to ^{sub space} $W \subset V$
 $W^\perp = [v \in V \text{ s.t. } \langle v, w \rangle = 0, \forall w \in W]$

* Defn
8.10 $v \in V$ is called a null vector if
 $\langle v, w \rangle = 0 \quad \forall w \in V$

* Defn
8.11 Nullspace N of $\langle \rangle$ is the subspace of null vectors.
 $N = [v \mid v \perp V] = V^\perp$

* Defn
8.12 $\langle, \rangle$ is called non degenerate on V if $N = \{0_v\}$

* Defn
8.13 $\langle, \rangle$ is called non degenerate on $W \subset V$
 if $W \cap W^\perp = \{0_v\}$
 another way of saying the same thing
 if $w \in W$ and $w \neq 0_v$ then $\exists w' \in W$ s.t. $\langle w, w' \rangle \neq 0$

* Caution A form may be non degenerate on the whole vector space and degenerate on subspace and vice versa.

* Lemma
8.14 $\langle, \rangle$ non degenerate $\iff$ ^{matrix of the form} A is invertible.

Chapter 8: Orthogonal basis

Saturday, 23 March 2024

* Theorem 8.15 $\langle, \rangle$ is non degenerate on $W \subset V$ if and only if $V = W \oplus W^\perp$

* Lemma 8.16 If $\langle, \rangle$ not identically zero $\exists v \in V$ s.t. $\langle v, v \rangle \neq 0$

* Theorem 8.17 Given $\langle, \rangle$ on V , there exists a orthogonal basis of V w.r.t $\langle, \rangle$

* Lemma 8.18 Given $\langle, \rangle$ on V , there exists a basis of V s.t the matrix of the form has this structure

$$A = \begin{bmatrix} I_m & & \\ & -I_n & \\ & & 0 \end{bmatrix}$$

* Defn 8.19 If $\langle, \rangle$ is non degenerate on $W \subset V$ then $V = W \oplus W^\perp$

Orthogonal Projection $\pi: V \longrightarrow V$ s.t

(linear transformation) $\pi(w) = w \quad \forall w \in W$

$\pi(w') = 0 \quad \forall w' \in W^\perp$

* Lemma 8.20 Projection Formula: If $[w_1, w_2, \dots, w_k]$ is orthogonal basis of W then

$$\pi(v) = c_1 w_1 + c_2 w_2 + \dots + c_k w_k$$

where

$$c_i = \frac{\langle v, w_i \rangle}{\langle w_i, w_i \rangle}$$

* Defn 8.21 Euclidean Space: Real Vector space V
+ a definite symmetric bilinear form

* Defn Hermitian Space: Complex Vector space V

* Defn
8.22

Hermitian Space: Complex Vector Space V
+ re definite Hermitian form

Chapter 8: Normal Operator

Saturday, 23 March 2024

* Lemma
8.23 If V is a Euclidean or Hermitian space. If $W \subset V$ then the form $\langle, \rangle$ is non degenerate on W and $W = W \oplus W^\perp$

* Defn
8.24 $T: V \rightarrow V$ be a linear operator on a Hermitian space.
 T is called
normal if its matrix A w.r.t a orthonormal basis has : $A^* A = A A^*$
hermitian $\gg$ " A " " " : $A^* = A$
Unitary $\gg$ " A " " " : $A^* A = I$

* Lemma
8.25 If A is a normal, hermitian or unitary then $P^* A P$ is also the same, where P is any unitary matrix.

* Defn
8.26 If $T: V \rightarrow V$ is a linear operator on Hermitian space
Adjoint operator $T^*: V \rightarrow V$ where the matrix of T^* is A^*

* Lemma
8.27 Let T be a linear operator on Hermitian space and T^* be the adjoint operator.
 $\forall v, w \in V$

- (a) $\langle T v, w \rangle = \langle v, T^* w \rangle$ and $\langle v, T w \rangle = \langle T^* v, w \rangle$
- (b) if T is normal $\langle T v, T w \rangle = \langle T^* v, T^* w \rangle$
- (c) if T is Hermitian $\langle T v, w \rangle = \langle v, T w \rangle$
- (d) if T is Unitary $\langle T v, T w \rangle = \langle v, w \rangle$

* Lemma
8.28 Let T be a linear operator on Hermitian space V . Let $W \subset V$
 W is T invariant $\iff W^\perp$ is T^* invariant

* Theorem If $T: V \rightarrow V$ normal operator on a Hermitian space V .
8.29 If (v, λ) is eigen vector of $T \rightarrow (v, \bar{\lambda})$ is eigenvector of T^*

Chapter 8: Spectral Theorem

Saturday, 23 March 2024

* Spectral Theorem 8.30: If $T: V \rightarrow V$ is a normal operator on a Hermitian Space. There is an orthonormal basis of V with eigen vectors of T .

* Lemma 8.31: If A is a normal matrix. There is unitary matrix P s.t P^*AP is diagonal

* Corollary 8.32: Every conjugacy class of a normal matrix contains a diagonal matrix

* Corollary 8.33: If $T: V \rightarrow V$ is an Hermitian operator on a Hermitian Space. Then the eigenvalues of T are real.

* Theorem 8.34: If $T: V \rightarrow V$ is an symmetric operator on a Euclidean Space. Then there is a orthonormal basis of V consisting of eigen vectors of T .

* Lemma 8.35: If A is a symmetric matrix then $\exists$ a orthonormal matrix P s.t P^TAP is diagonal

* Defn 8.36: A bilinear form $\langle, \rangle$ on a vector space is called skew-symmetric if $\langle u, v \rangle = 0 \quad \forall u, v \in V$ — (1)

$$\text{or} \quad \langle u, v \rangle = -\langle v, u \rangle \quad \forall u, v \in V \quad \text{--- (2)}$$

These conditions are same if the characteristic of F is not 2.

If char 2 condition (1) is correct

* Lemma 8.37: If $\langle, \rangle$ is a non degenerate skew symmetric form on vector space V the $\exists$ a basis $[B]$ of V s.t the matrix of the form has the following structure.

$$\begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix}$$

$$\begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix}$$

2.57 the $\mathcal{F}$ a basis $\{B\}$ of V s.t the matrix of the form has the following structure.

$$S_0 = \begin{bmatrix} \mathcal{E} & & \\ & \mathcal{E} & \\ & & \ddots \end{bmatrix} \quad \text{where } \mathcal{E} = \begin{bmatrix} -1 & 1 \end{bmatrix}$$

Chapter 11: Rings

Saturday, 23 March 2024

* Defn Ring $(+, -, \times, 1)$: A ring is a set R together with two laws of composition $+, \times$ s.t

- ① R is an abelian group under $+$, identity 0_R
- ② R is closed under $\times$, $\times$ is commutative, $\nexists$ identity 1
- ③ $\forall a, b, c \in R$
$$a \times (b + c) = a \times b + a \times c$$

Example: Subrings of $\mathbb{C}$

$$\begin{aligned} &\rightarrow \mathbb{Z} \\ &\rightarrow \mathbb{Z}[\alpha] = \text{Smallest subring of } \mathbb{C} \text{ that contains } \alpha \\ &\quad = \left\{ \beta \in \mathbb{C} \text{ s.t.} \right. \\ &\quad \quad \left. \beta = a_1 \alpha^1 + a_2 \alpha^{2-1} + \dots + a_0 \text{ where} \right\} \\ &\quad \quad \quad a_i \in \mathbb{Z} \\ &\rightarrow \mathbb{Z}[i] = \{a + bi \text{ where } a, b \in \mathbb{Z}\} \end{aligned}$$

* Defn Unit: Elements of ring that have multiplicative inverses

* Defn $R[x] =$ set of polynomials with co-efficients from ring R

$$R[x] = \left\{ f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0 \right\} \\ \text{where } a_i \in R$$

* Lemma If R is a ring, $R[x]$ is also a ring. R becomes a subring of $R[x]$ where its elements are identified with constant polynomials

* Lemma Let R be a ring and $f(x), g(x) \in R[x]$ s.t $f(x)$ is monic polynomial, i.e. its leading co-efficient is identity, then there are unique $q(x), r(x) \in R[x]$ s.t
$$g(x) = f(x) q(x) + r(x)$$

$q(x), r(x) \in \mathbb{R}[x]$ s.t

$$g(x) = q(x) f(x) + r(x)$$

where $\deg(r) < \deg f(x)$ or $r=0$

Chapter 11: Substitution Principle

Saturday, 23 March 2024

* Defn Ring homomorphism: $\varphi: R \rightarrow R'$
 s.t.

- $\varphi(a+b) = \varphi(a) + \varphi(b) \quad \forall a, b \in R$
- $\varphi(ab) = \varphi(a) \cdot \varphi(b) \quad \forall a, b \in R$
- $\varphi(1_R) = 1_{R'}$

$$\text{kernel}(\varphi) = \{a \in R \text{ s.t. } \varphi(a) = 0_{R'}\}$$

└ closed under addition

└ closed under multiplication within R ie

if $a \in R$ $b \in \text{kernel}(\varphi)$ then $a \cdot b \in \text{kernel}(\varphi)$

* Lemma (Substitution Principle)
 If $\varphi: R \rightarrow R'$ is a ring homomorphism, $\exists \phi: R[x] \rightarrow R'$, a ring homomorphism, s.t.

- ϕ acts as φ on $R \subset R[x]$
- $\phi: x \mapsto \alpha$ where $\alpha \in R'$

 Caution: α need not to be $\in \text{Image}(\varphi)$

Example: Let $\varphi: \mathbb{Z} \rightarrow \mathbb{C}$ be the inclusion homomorphism

$\phi: \mathbb{Z}[x] \rightarrow \mathbb{C}$ s.t. $x \mapsto \alpha$ is an homomorphism

if α is not algebraic ie not the root of a polynomial with integer coefficients

ϕ is injective

Example: Let $\varphi: \mathbb{Z} \rightarrow \mathbb{Z}[x][y]$ be the inclusion homomorphism

$\phi: \mathbb{Z}[x, y] \rightarrow \mathbb{Z}[x][y]$ is an unique isomorphism

ie. $\mathbb{Z}[x, y]$ is isomorphic to $\mathbb{Z}[x][y]$

* Lemma $\varphi: \mathbb{Z} \rightarrow R$

unique and is defined $\varphi(n) = \underbrace{1_R + 1_R + \dots + 1_R}_{n \text{ terms}}$

Chapter 11: Ideal

Saturday, 23 March 2024

* Defn Ideal: An Ideal I of a ring R is a non empty subset s.t

- ① Set is closed under addition within I
- ② Set is closed under multiplication within R , i.e.
 $\forall a \in I, b \in R \quad a \cdot b \in I$

* Defn An Ideal generated by a set $S = \{s_1, s_2, \dots, s_k\}$ is the smallest Ideal that contains these elements.

$$I = \left\{ x_1 s_1 + x_2 s_2 + \dots + x_k s_k \text{ where } \right. \\ \left. x_i \in R \text{ and } s_i \in S \right\}$$

* Defn Principal Ideal: An Ideal generated by a single element $a \in R$ is called principal Ideal. It is denoted by

$$(a) = aR = Ra = \{xa \text{ where } x \in R\}$$

* Lemma An field has only two Ideals (0) and (1) $\iff$ A ring that has exactly two Ideals is a field.

*